

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	DSA0307	Course Title:	Natural Language Processing
CO Assessed:	CO2 – Manipulation	Assessment:	Assessment Tool 1 – ANALYTICAL PROBLEM SOLVING
Weightage:	50% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
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Section / Batch:	B.E-CSE-AI	Date:	

Code of Conduct

I Shaik Nuha (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sk.Nuha.

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Part of Speech Tagging	Morphological decomposition of connected, connecting, and connection; identification of roots, suffixes, inflectional vs. derivational forms, and normalization for document indexing.	Q1
English Word Classes, Tagsets for English, Rule-Based Part of Speech Tagging	Morphological parsing of unhappy, happiness, and happily; identification of prefixes, suffixes, base forms, and semantic effects of derivational morphology.	Q2
Counting Words, Part of Speech Tagging, English Word Classes	Stemming and root extraction for played, player, and playing; handling grammatical variations and word formation changes for search preprocessing.	Q3

Finite-State Morphological Analysis, Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Finite-state parsing of writes, writing, and written; modeling regular and irregular verb inflections and root normalization.	Q4
English Word Classes, Rule-Based Part of Speech Tagging, Counting Words	Porter Stemmer-based processing of relational, relation, and relate; application of derivational suffix removal and stem extraction for retrieval.	Q5

Annexure A – ANALYTICAL PROBLEM SOLVING

1. Given the input words: “connected”, “connecting”, “connection”, design a morphological analysis pipeline for a document indexing system.

Apply rules to decompose each word into root and suffix components.

Differentiate between inflectional and derivational forms within the dataset.

Normalize all variants to a common base form for efficient retrieval.

Determine the parsed structure and normalized output for each word.

2. Given the input words: “unhappy”, “happiness”, “happily”, design a morphological parsing module for sentiment analysis.

Identify prefixes, suffixes, and base forms using rule-based decomposition.

Ensure semantic consistency across derived word forms.

Classify each transformation as inflectional or derivational.

Produce the root and morphological breakdown for each word.

3. Given the input words: “played”, “player”, “playing”, design a stemming-based preprocessing module for a search engine.

Apply morphological rules to strip affixes and identify the base form.

Handle both grammatical variations and word formation changes.

Ensure consistent root extraction across all forms.

Determine the stem and classification for each input word.

4. Given the input words: “writes”, “writing”, “written”, design a finite-state morphological parsing module for a text processing system.

Apply state transitions to represent suffix transformations and irregular forms in verb inflection.

Differentiate between regular and irregular inflectional patterns within the dataset.

Ensure consistent mapping of all forms to a common root representation.

Determine the state transitions, morphological breakdown, and normalized output for each word.

5. Given the input words: “relational”, “relation”, “relate”, design a Porter Stemmer-based preprocessing module for document retrieval.

Apply stemming rules step-by-step to remove derivational suffixes and obtain base forms.

Handle variations in suffix patterns while preserving semantic consistency.

Ensure uniform root extraction across all derived forms.

Determine the stemming steps, intermediate forms, and final stem for each input word.

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Q. No. / Criteria Level	Problem Understanding	Method Selection	Solution Process	Accuracy of Calculation	Interpretation of Results	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1. Morphological Analysis of "talked", "talking", and "talks"

Aim: To apply inflectional morphology rules and Porter Stemmer to derive the normalized root form and explain how morphological normalization improves information retrieval.

Step 1: Inflectional Morphology

Word	Root	Suffix Removed	Type
talked	talk	-ed	Past Tense
talking	talk	-ing	Present Participle
talks	talk	-s	Plural/Third Person

Step 2: Porter Stemmer

Word	Porter Stem	Rule Applied
talked	talk	Remove "ed"
talking	talk	Remove "ing"
talks	talk	Remove "s"

Final Normalized Form

talk

Morphological Normalization

Without stemming:

talked

talking

talks

are considered different words.

After normalization:

talk

All words are indexed under one root.

Advantages:

- Reduces redundant words
- Improves search accuracy
- Faster indexing
- Better text mining
- Better document retrieval

Conclusion:

Inflectional morphology removes grammatical endings, while Porter Stemmer normalizes all word forms into the common root talk, improving Information Retrieval accuracy.

2. Hybrid Morphological Processing Pipeline

Aim: To design a hybrid morphological processing pipeline using

- Derivational Morphology
- Inflectional Morphology
- Finite State Morphological Parsing
- Porter Stemmer

Step 1: Morphological Decomposition

Word	Root	Affix	Type
relational	relate	-ional	Derivational
conditional	condition	-al	Derivational
happiness	happy	-ness	Derivational
running	run	-ing	Inflectional

Step 2: Finite State Parsing

relational

Start

↓

relate

↓

+ ion

↓

+ al

↓

Accept

conditional

Start



condition



+ al



Accept

happiness

Start



happy



+ ness



Accept

running

Start



run



+ ing



Accept

Step 3: Porter Stemmer

Word	Stem
relational	relat
conditional	condit
happiness	happi
running	run

Final Normalized Forms

Word	Final Stem
relational	relat
conditional	condit
happiness	happi
running	run

Contribution of Each Technique:

1.Derivational Morphology

Creates new words by adding suffixes.

Example:

happy → happiness

2.Inflectional Morphology

Changes grammatical form.

Example:

run → running

3.Finite State Morphology

Validates legal word formation using state transitions.

4.Porter Stemmer

Produces one common searchable root.

Conclusion:

The hybrid pipeline accurately extracts roots and normalizes related words, improving NLP applications such as search engines, indexing, and text mining.

3. Morphology-driven SEO System

Aim: To develop a morphology-driven SEO system integrating

- Inflectional Analysis
- Derivational Analysis
- Finite State Parsing
- Porter Stemmer

for consistent keyword indexing.

Input Words

optimize

optimization

optimized

optimizing

Step 1: Morphological Analysis

Word	Root	Affix	Type
optimize	optimize	—	Base Word
optimization	optimize	-ation	Derivational
optimized	optimize	-ed	Inflectional
optimizing	optimize	-ing	Inflectional

Step 2: Finite State Parsing

optimize

Start

↓

optimize

↓

Accept

optimization

Start

↓

optimize

↓

+ ation

↓

Accept

optimized

Start

↓

optimize

↓

+ ed

↓

Accept

optimizing

Start

↓

optimize

↓

+ ing
↓
Accept

Step 3: Porter Stemmer

Word	Stem
optimize	optim
optimization	optim
optimized	optim
optimizing	optim

Final Normalized Form

optim

SEO Benefits

- Removes duplicate keywords
- Improves keyword indexing
- Better page ranking
- Reduces storage redundancy
- Improves search relevance

Working of the Hybrid Pipeline

Input Words



Morphological Analysis



Inflectional Analysis



Derivational Analysis



Finite State Parsing



Porter Stemmer



Normalized Root



Search Engine Index

Conclusion:

The hybrid morphology-driven SEO system converts optimize, optimization, optimized, and optimizing into the common stem optim. This enables consistent indexing, reduces redundancy, and improves search relevance and ranking.

4. Hybrid Morphological Processing for "compute", "computer", "computing", and "computation"

Aim: To design a hybrid morphological processing pipeline integrating inflectional analysis, derivational analysis, finite-state morphological parsing, and Porter Stemmer to normalize different word forms for efficient information retrieval.

Step 1: Morphological Decomposition

Word	Root	Affix	Type
compute	compute	—	Base Word
computer	compute	-er	Derivational
computing	compute	-ing	Inflectional
computation	compute	-ation	Derivational

Step 2: Finite-State Morphological Parsing

compute

Start



compute



Accept

computer

Start



compute



+ er



Accept

computing

Start



compute



+ ing



Accept

computation

Start



compute



+ ation



Accept

Step 3: Porter Stemmer

Word	Stem	Rule Applied
compute	comput	Remove final 'e'
computer	comput	Remove "-er"
computing	comput	Remove "-ing"
computation	comput	Remove "-ation"

Final Normalized Forms

Word	Final Stem
compute	comput
computer	comput
computing	comput
computation	comput

Hybrid Morphological Pipeline

Input Words



Derivational Analysis



Inflectional Analysis



Finite-State Parsing



Porter Stemmer



Normalized Root



Information Retrieval Index

Contribution of Each Technique:

1. Inflectional Analysis

Removes grammatical endings like -ing.

2. Derivational Analysis

Identifies new words formed using suffixes such as -er and -ation.

3. Finite-State Morphological Parsing

Validates word formation using state transitions.

4. Porter Stemmer

Produces a common searchable root (comput).

Advantages:

- Consistent indexing
- Improved search accuracy
- Reduced redundancy
- Faster document retrieval
- Better text mining performance

Conclusion:

The hybrid morphological pipeline successfully extracts the common root comput from all word variations, ensuring efficient indexing and improved search accuracy.

5. Morphology-Based Framework for "decide", "decision", "decisive", and "deciding"

Aim: To construct a morphology-based framework combining derivational rules, inflectional transformations, finite-state parsing, and Porter Stemming for semantic clustering and text analysis.

Step 1: Morphological Decomposition

Word	Root	Affix	Type
decide	decide	—	Base Word
decision	decide	-sion	Derivational
decisive	decide	-ive	Derivational
deciding	decide	-ing	Inflectional

Step 2: Finite-State Parsing

decide

Start



decide



Accept

decision

Start



decide



+ sion



Accept

decisive

Start



decide



+ ive



Accept

deciding

Start



decide



+ ing



Accept

Step 3: Porter Stemmer

Word	Stem	Rule Applied
decide	decid	Remove final 'e'
decision	decis	Remove "-sion"
decisive	decis	Remove "-ive"
deciding	decid	Remove "-ing"

Final Normalized Forms

Word	Final Stem
decide	decid
decision	decis
decisive	decis
deciding	decid

Hybrid Morphological Pipeline:

Input Words



Derivational Analysis



Inflectional Analysis



Finite-State Parsing



Porter Stemmer



Normalized Root



Semantic Clustering

Effect of Suffix Variations

Suffix	Effect
-sion	Converts a verb into a noun
-ive	Converts a verb into an adjective
-ing	Represents a continuous verb form

Although the grammatical category changes, all words originate from the same base concept decide, helping maintain semantic consistency.

Contribution of Each Technique:

1.Derivational Morphology

Creates new words with different meanings and grammatical categories.

2.Inflectional Morphology

Changes the grammatical form without changing the core meaning.

3.Finite-State Morphological Parsing

Ensures valid word formation using state transitions.

4.Porter Stemmer

Normalizes related words to a common stem for efficient indexing.

Advantages:

- Improves semantic clustering
- Reduces duplicate terms
- Enhances search relevance
- Supports efficient text mining
- Improves NLP preprocessing

Conclusion:

The morphology-based framework successfully analyzes decide, decision, decisive, and deciding, extracting common roots and validating transformations. By combining derivational and inflectional morphology, finite-state parsing, and Porter Stemming, the system improves semantic consistency, indexing efficiency, and overall NLP performance.